D-modules
Lectures by Victor Ginzburg

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Preface

There will be some gaps in explanation, either due to the lecturer's admission or my own lack of understanding. In particular, many "proofs" are sketches of proofs. Gaps due to my own misunderstanding will be indicated by three red question marks: ???. More generally, my own questions about the material will also be in red. Things like "**Question**" will be questions posed by the lecturer. Feel free to reach out to me with explanations.

Contents

1 Mar 24 **3**
1.1 Applications of D-modules 4
1.2 Nearby cycles 5
1.3 Microlocal geometry 5
1.4 de Rham functor 5

2 Mar 26 **6**
2.1 Poincaré-Birkhoff-Witt (PBW) for differential operators 6
2.2 An application to the PBW theorem for Lie algebras 7

3 Mar 31 **9**
3.1 Sheaves of differential operators 9
3.2 Twisted differential operators (TDO) 10
3.3 Beilinson-Bernstein localization theorem, the statement 11

4 Apr 2 **12**
4.1 Beilinson-Bernstein localization theorem, the proof (part 1) . . . 12
4.1.1 Setup 12
4.1.2 Step 1 12
4.1.3 Step 2 12
4.1.4 Step 3 13

4.1.5 Step 4 13
4.1.6 Step 5 14
4.1.7 Step 6 14
4.2 Some setup for next time 14

1 Mar 24

D-modules are modules over rings of differential operators. There are three settings:

1. C^∞ case: We can consider differential operators on $\mathbb{R}^n$ of order N, which are expressions

$$u = \sum_{i_1+\cdots+i_n \le N} a_{i_1,\ldots,i_n}(x) \frac{\partial^N}{\partial x_1^{i_1} \cdots \partial x_n^{i_n}},$$

   where the functions $a_\bullet(x)$ are smooth.

2. Complex analytic setting: Similar, but replace $\mathbb{R}$ with $\mathbb{C}$ and require the functions $a_\bullet(x)$ are holomorphic.

3. Algebraic setting: Replace $\mathbb{C}$ with any field and require the functions $a_\bullet(x)$ to be polynomials.

We will work with smooth algebraic varieties over $\mathbb{C}$.

Remark. 1. If X is singular, the ring $\mathcal{D}(X)$ of differential operators might not be noetherian.

2. The theory of differential operators in positive characteristic is very different.

3. We can effectively replace $\mathbb{C}$ by any algebraically closed field of characteristic zero.

Today X is smooth and affine. $\mathcal{O} = \mathcal{O}(X)$ will denote the algebra of regular functions on X. We might not have coordinates on X, so we need to modify our definition of differential operators.

Differential operators of order 0 are just multiplication by some $f \in \mathcal{O}$, denoted m_f. Differential operators of order 1 are $\xi + m_f$, where $f \in \mathcal{O}$ and ξ is a vector field on X, i.e. an element of $T_X = \mathrm{Der}(\mathcal{O}, \mathcal{O})$, i.e. a derivation $\mathcal{O} \to \mathcal{O}$.

Definition 1.1. The algebra $\mathcal{D} = \mathcal{D}(X)$ of differential operators on X is defined as the subalgebra of $\mathrm{End}_{\mathbb{C}}(\mathcal{O})$ generated by the operators of order 0 and 1 defined above.

Note that $\mathcal{O} + T_X$ sits as a subspace in $\mathcal{D}$.

When $X = \mathbb{C}^n$, or more generally is an open subvariety of $\mathbb{C}^n$, then this definition agrees with the previous one using the induced coordinates from $\mathbb{C}^n$.

1.1 Applications of D-modules

Fix $f \in \mathcal{O}$. Let $\mathrm{Ann}(f) = \{u \in \mathcal{D} \mid u(f) = 0\}$. It is a left ideal in $\mathcal{D}$. We define $\mathcal{D}f$ to be $\mathcal{D}/\mathrm{Ann}(f)$, which is a $\mathcal{D}$-module.

We can define $\mathcal{D}f$ for some $f \notin \mathcal{O}$:

1. Let $X = \mathbb{C}$ and $f = e^x$. Certainly f is not regular, but its annihilator is generated by $\frac{d}{dx} - 1$, so we can define $\mathcal{D}e^x = \mathcal{D}/(\frac{d}{dx} - 1)$.

2. Similarly we can define $\mathrm{Ann}(f)$ and hence $\mathcal{D}f$ for $X = \mathbb{C}$ and $f \in C^\infty(\mathbb{R})$.

3. Let $f \in \mathbb{R}[x_1, \ldots, x_n]$ and let $U = \{x \in \mathbb{R}^n \mid f(x) \geq 0\}$. Fix $\varphi \in C^\infty(U)$ with compact support. Let $I_\phi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx$ for $\lambda \in \mathbb{C}$. Note that the exponentiation is well-defined because $f(x) \geq 0$. The integral converges when the real part of λ is positive. Then

   Theorem 1.1. *The function $\lambda \mapsto I_\varphi(\lambda)$ extends to a meromorphic function on $\mathbb{C}$ with poles at a finite union of arithmetic progressions of the form $a, a-1, \ldots$.*

   Sketch. Say $I_\varphi(\lambda)$ converges for $\lambda > N$ (at the start, we have $N = 0$, but we proceed inductively). We compute

   $$\frac{d}{d\lambda} I_\varphi(\lambda) = \int_U \varphi(x) \log f(x) f(x)^\lambda dx,$$

   which converges, so $I_\varphi(\lambda)$ is holomorphic when $\mathrm{Re}\lambda \geq N$. Our goal is to extend the function to $\mathrm{Re}\lambda \geq N - 1$ inductively. We need:

   Lemma 1.1 (b-function)**.** *There exists $u \in \mathcal{D}(\mathbb{C}^n)[s]$ and $b \in \mathbb{C}[s]$ such that $u(f^{s+1}) = b(s) f^s$.*

   Assuming the lemma, we have

   $$\begin{aligned} I_\varphi(\lambda) = \int_U \varphi(x) f(x)^\lambda dx &= \int_U \varphi(x) \frac{1}{b(\lambda)} u(f(x)^{\lambda+1}) dx \\ &= \frac{1}{b(\lambda)} \int_U (u^*(\varphi))(x) f(x)^{\lambda+1} dx, \end{aligned}$$

   where at the end we have used essentially integration by parts, and $u^* \in \mathcal{D}(\mathbb{C}^n)[s]$ is the formal adjoint of u. For $\mathrm{Re}\lambda > N - 1$ we can extend $I_\varphi(\lambda)$ using the above computation.

Example 1.1 (Examples of b-functions)**.**

1. Let $f(x) = \sum x_i^2$ on $\mathbb{C}^n$. Then we can take $u = \sum \frac{\partial^2}{\partial x_i^2}$ and $b(s) = 4(s+1)(s+\frac{n}{2})$.

2. Let $f(x) = \det(x)$ on $\mathbb{C}^{n^2} = M_n(\mathbb{C})$. Then for $u = \det(\frac{\partial}{\partial x_{ij}})$ one can compute $u(f^s)$ using Capelli identities and find that $b(s) = \prod_{1 \leq k \leq n} (s+k)$.

3. If $f(x_1, \ldots, x_n) = \prod_{1 \le i < j \le n}$, then the b-function is unknown. Can also generalize to arbitrary root systems.

4. If one takes the square of the above function, then the b-function is known, but is very complicated.

1.2 Nearby cycles

Let $f \in \mathbb{C}[x_1, \ldots, x_n]$. Let $\mathrm{Sing}(f) = \{x \in \mathbb{C}^n \mid df(x) = 0\}$. Fix $z_0 \in \mathrm{Sing}(f)$. Take $0 < \delta \ll \varepsilon \ll 1$. We define the *link* of f at z_0 to be

$$L = L(f, z_0, \varepsilon, \delta) = \{z \in \mathbb{C}^n \mid \|z - z_0\| \le \varepsilon, \|f(z) - f(z_0)\| = \delta\}.$$

Topologically, L is independent of δ, ε. there's a nice picture but seems hard to TiKz. Let S^1_δ denote the circle of w such that $\|w - f(z_0)\| = \delta$. The family $H_p(L_w)$ forms a locally constant sheaf on S^1_δ. For fixed w, there is a *monodromy operator* $M_{f,p} : H_p(L_w) \to H_p(L_w)$.

Theorem 1.2. *If $b_f(\lambda) = 0$, then $e^{2\pi i \lambda}$ is an eigenvalue of $M_{f,p}$ for some p. Moreover if $\mathrm{Sing}(f) = \{z_0\}$, then all eigenvalues of all $M_{f,p}$ arise in this way.*

1.3 Microlocal geometry

$\mathcal{D}(X)$ is a noncommutative deformation (aka quantization) of $\mathcal{O}(T^*X)$.

For $f \in \mathcal{O}(X)$, we have $\mathrm{Sing}(f) \subset X$. We also have the $\mathcal{D}$-module $\mathcal{D}f$. To a $\mathcal{D}$-module we can define its singular support, which sits in T^*X. The following square commutes:

$$\begin{tikzcd} SS(\mathcal{D}f) \arrow[r, hook] \arrow[d] & T^*X \arrow[d] \\ \mathrm{Sing}(f) \arrow[r, hook] & X \end{tikzcd}$$

The singular support of $\mathcal{D}f$ can contain more information than just $\mathrm{Sing}(f)$.

1.4 de Rham functor

There is a functor from the (derived?) category of $\mathcal{D}(X)$-modules to the (derived?) category of complexes of sheaves of $\mathbb{C}$-vector spaces on X. It restricts to a functor between the (derived?) category of holonomic modules to the category of perverse sheaves. This further restricts to an equivalence of categories between holonomic modules with regular singularities (or complexes with regular holonomic cohomology?) to the category of perverse sheaves.

2 Mar 26

2.1 Poincaré-Birkhoff-Witt (PBW) for differential operators

We continue to assume X is smooth affine, and we write $\mathcal{O}, \mathcal{D}$ for $\mathcal{O}(X), \mathcal{D}(X)$. We let T_X denote the space of vector fields.

For $\xi \in T_X$ and $f, g \in \mathcal{O}$, we have the Liebniz formula $\xi(fg) = f\xi(g) + \xi(f)g$. In particular, in terms of operators, $\xi m_f - m_f \xi = m_{\xi(f)}$. Also, for $\eta \in T_X$, we have $\xi\eta - \eta\xi = [\xi, \eta]$, where $[-,-]$ is the (a priori abstract) Lie bracket of vector fields. It follows that $\mathcal{D}_1 = \mathcal{O} + T_X$ is a Lie algebra. $\mathcal{O}$ is an abelian Lie ideal and T_X is a Lie subalgebra. We have a short exact sequence $0 \to \mathcal{O} \to \mathcal{D}_1 \to T_X \to 0$. The formula $\xi m_f - m_f \xi = m_{\xi(f)}$ also shows that $\mathcal{D}_1$ is an $\mathcal{O}$-bimodule, and that the left and right actions agree on $\mathcal{D}_1/\mathcal{O}$.

Definition 2.1. The **order filtration** on $\mathcal{D}$ is given by $\mathcal{D}_0 = \mathcal{O}$, $\mathcal{D}_1 = \mathcal{O} + T_X$, and $\mathcal{D}_n$ is the image of the composition map $\mathcal{D}_1 \otimes_{\mathcal{O}} \cdots \otimes_{\mathcal{O}} \mathcal{D}_1 \to \mathcal{D}$.

We have

- $\bigcup_n \mathcal{D}_n = \mathcal{D}$.
- $\mathcal{D}_n$ is an $\mathcal{O}$-bimodule.
- $\mathcal{D}_m \mathcal{D}_n \subset \mathcal{D}_{m+n}$.

Let $\mathrm{gr}\mathcal{D} = \bigoplus(\mathcal{D}_n/\mathcal{D}_{n-1})$. Let $f \in \mathcal{O}$ and $\xi, \eta \in T_X$. Then $f\xi \equiv \xi f \mod \mathcal{O}$, so $\mathcal{O} = \mathcal{D}_0$ is a central subalgebra of $\mathrm{gr}\mathcal{D}$. Similarly, $\xi\eta \equiv \eta\xi \mod \mathcal{D}_1$, so $T_X = \mathcal{D}_1/\mathcal{D}_0$ is contained in $Z(\mathrm{gr}\mathcal{D})$. We obtain a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathcal{O}} T_X \to \mathrm{gr}\mathcal{D}$.

Theorem 2.1 (PBW for $\mathcal{D}$)**.** *μ is an isomorphism.*

Proof. It suffices to show that μ is injective.

The cotangent bundle T^*X is a vector bundle, so it admits an (attracting) algebraic $\mathbb{C}^\times$-action given by fiberwise scaling. This gives a grading $\mathcal{O}(T^*X) = \bigoplus_{n \geq 0} \mathcal{O}_n(T^*X)$, where $\mathcal{O}_n(T^*X)$ consists of functions that are fiberwise homogeneous of degree n. Now, there is a canonical isomorphism $\nu : \mathrm{Sym}^n_{\mathcal{O}}(T_X) \xrightarrow{\sim} \mathcal{O}_n(T^*X)$, where a product of vectors $\xi_1 \cdots \xi_n$ is sent to the function taking $(x, \alpha) \in T^*X$ to $\prod \langle (\xi_i)_x, \alpha \rangle$.

Now, there is a so-called principal symbol map that we will define in a moment, $\sigma_n : \mathcal{D}_n \to \mathcal{O}_n(T^*X)$, that sends $\mathcal{D}_{n-1}$ to 0, so it descends to a map $\overline{\sigma}_n : \mathcal{D}_n/\mathcal{D}_{n-1} \to \mathcal{O}_n(T^*X)$. We will check that the composite

$$\mathrm{Sym}^n_{\mathcal{O}} T_X \xrightarrow{\mu} \mathcal{D}_n/\mathcal{D}_{n-1} \xrightarrow{\overline{\sigma}_n} \mathcal{O}_n(T^*X) \xrightarrow{\nu^{-1}} \mathrm{Sym}^n_{\mathcal{O}} T_X$$

is the identity, which implies μ is injective.

Now we construct σ_n. Note that $\mathcal{D}$ acts on $\mathcal{O}[[t]]$ by $u(\sum f_i t^i) = \sum u(f_i)t^i$. For fixed $f \in \mathcal{O}$, we have $e^{tf} = \sum \frac{1}{i!} f^i t^i \in \mathcal{O}[[t]]$.

Claim 2.1. *1. For $u \in \mathcal{D}_n$, we have*

$$u(e^{tf}) = (t^n \sigma_n(u, f) + \text{ lower degree terms})\, e^{tf},$$

where $\sigma_n(u, f), A \in \mathcal{O}$.

2. If $u \in \mathcal{D}_{n-1}$, then $\sigma_n(u, f) = 0$.

3. We have

$$\sigma_n(\xi_1 \cdots \xi_n, f)(x) = \prod \langle (\xi_i)_x, d_x f \rangle = \nu(\xi_1 \cdots \xi_n)(d_x f).$$

Proof of Claim. First we have

$$\xi e^{tf} = t\xi(f)e^{tf} = t\langle \xi, df \rangle e^{tf}.$$

Then we have

$$\xi_1\xi_2 e^{tf} = t^2 \xi_1(f)\xi_2(f)e^{tf} + t\xi_1\xi_2(f)e^{tf} = \left(t^2 \langle \xi_1, df \rangle \langle \xi_2, df \rangle + t\xi_1\xi_2(f)\right) e^{tf}.$$

And from here the pattern is clear.

This claim both gives the construction of σ_n and shows that the previous composition is the identity, so we are done. I want to check this in more detail I think

Remark. PBW and its proof work in the C^∞ or complex analytic settings.

2.2 An application to the PBW theorem for Lie algebras

Let $\mathfrak{g}$ be a Lie algebra, and let $\mathcal{U}\mathfrak{g}$ be its universal enveloping algebra. Lecturer: "since this is not a course on Lie algebras, I'm not supposed to explain what it is." $\mathcal{U}\mathfrak{g}$ has a PBW filtration defined in a very similar way to the order filtration. In particular $\mathcal{U}_0\mathfrak{g} = \mathbb{C}$ and $\mathcal{U}_1\mathfrak{g} = \mathbb{C} + \mathfrak{g}$. Furthermore, the relations of $\mathcal{U}\mathfrak{g}$ imply that we get a surjective graded algebra map $\mu : \mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$. The PBW theorem (for Lie algebras) asserts that μ is an isomorphism. It is not easy, but if we assume that $\mathfrak{g}$ is the Lie algebra of a Lie group or affine algebraic group $\mathfrak{g}$, then it is easy.

Note. It is in fact true that every $\mathfrak{g}$ comes from a G, but the proof is no easier than PBW, and in some cases one uses PBW in the proof.

We can identify $\mathfrak{g}$ with $(T_G)^G$, i.e. left-invariant vector fields on G, as well as with T_eG. (At least one of these is the definition of Lie$G$, depending on your conventions.) We get a map $\mathfrak{g} \to \mathcal{D}(G)^G$ by sending a vector field to its corresponding differential operator, which gives a map $\tau : \mathcal{U}\mathfrak{g} \to \mathcal{D}(G)^G$.

Theorem 2.2. *1. The map τ is an algebra isomorphism.*

2. PBW for $\mathcal{U}\mathfrak{g}$ is true.

Proof. By construction, $\tau(\mathcal{U}_n\mathfrak{g}) \subset \mathcal{D}_n(G)^G$, so there is an induced graded map $\mathrm{gr}(\tau) : \mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$. We then have the following commutative diagram:

$$\begin{tikzcd}
\mathrm{Sym}^n_{\mathbb{C}}\mathfrak{g} \arrow[r, "\sim"] \arrow[d, two heads, "\mu_n"'] & \mathrm{Sym}^n_{\mathcal{O}}((T_G)^G) \arrow[r, "\sim"] & (\mathrm{Sym}^n_{\mathcal{O}}T_G)^G \arrow[d, "\sim\text{by PBW for } \mathcal{D}"] \\
\mathcal{U}_n\mathfrak{g}/\mathcal{U}_{n-1}\mathfrak{g} \arrow[r, "\mathrm{gr}_n(\tau)"'] & \mathcal{D}_n(G)^G/\mathcal{D}_{n-1}(G)^G \arrow[r, hook] & (\mathcal{D}_n(G)/\mathcal{D}_{n-1}(G))^G
\end{tikzcd}$$

The right map on the bottom is injective because $\mathcal{D}_n(G)^G \cap \mathcal{D}_{n-1}(G) = \mathcal{D}_{n-1}(G)^G$. It follows that all arrows are isomorphisms, so we get that $\mathrm{Sym}_{\mathbb{C}}\mathfrak{g} \to \mathrm{gr}\mathcal{U}\mathfrak{g}$ and $\mathrm{gr}\mathcal{U}\mathfrak{g} \to \mathrm{gr}\mathcal{D}(G)^G$ are both isomorphisms. The former isomorphism is PBW, and the latter isomorphism implies τ is an isomorphism, so we are done.

We now discuss another formulation of $\mathcal{D}(X)$. By the lecturer's admission, this may be wrong as it was done on the spot. We have $\mathcal{D}(X) = \mathcal{U}_{\mathbb{C}}(\mathcal{O}+T_X)/(f\otimes\xi = f\xi; 1_{\mathcal{D}} = 1_{\mathcal{O}} = 1_{\mathcal{U}})$. The proof uses PBW to show that the associated graded algebras agree.

3 Mar 31

3.1 Sheaves of differential operators

Let X be an irreducible affine. Then $\mathcal{O}(X)$ is a domain. For any nonzero $f \in \mathcal{O}(X)$, we have the basic open $X_f = \{x \in X \mid f(x) \neq 0\}$. It is affine with coordinate ring $\mathcal{O}(X)_f = \mathcal{O}(X)[\frac{1}{f}]$.

Now assume X is not necessarily affine. Facts:

1. A sheaf $\mathcal{E}$ of $\mathcal{O}_X$-modules is quasicoherent if and only if for every open affine $U \subset X$ and nonzero $f \in \mathcal{O}(U)$, the map

$$\Gamma(U, \mathcal{E})_f := \mathcal{O}(U_f) \otimes_{\mathcal{O}(U)} \Gamma(U, \mathcal{E}) \to \Gamma(U_f, \mathcal{E}) \tag{1}$$

   is an isomorphism. In this case/equivalently, for all affine opens $V \subset U$, there is an isomorphism

$$\mathcal{O}(V) \otimes_{\mathcal{O}(U)} \Gamma(U, \mathcal{E}) \to \Gamma(V, \mathcal{E}).$$

2. If X is affine, then the map

$$\mathrm{Sym}^n_{\mathcal{O}(X)} \Gamma(X, \mathcal{E}) \to \Gamma(X, \mathrm{Sym}^n_{\mathcal{O}_X} \mathcal{E})$$

   is an isomorphism.

Continue assuming X is arbitrary. Suppose to every open affine $U \subset X$ we associate an $\mathcal{O}(U)$-module $\mathcal{E}(U)$, such that there are compatible restriction maps $\mathrm{res} : \mathcal{E}(U) \to \mathcal{E}(U_f)$. For $x \in X$ we define $\mathcal{E}_x = \varinjlim \mathcal{E}(U)$, with the colimit taken over open affines U containing x. This gives a sheaf of $\mathcal{O}_X$ modules that we denote by $\mathcal{E}$. If the map 1 is an isomorphism, then $\mathcal{E}$ is quasicoherent. If X is affine, then there is a map $\mathcal{E}(X) \to \Gamma(X, \mathcal{E})$. If the map 1 is an isomorphism, then so is $\mathcal{E}(X) \to \Gamma(X, \mathcal{E})$.

Let T_X be the tangent sheaf on X. If X is affine, we put $\mathrm{Vect}(X) = \Gamma(X, T_X)$. We define $\mathrm{Sym}^n \mathrm{Vect}(X)$ to be $\mathrm{Sym}^n_{\mathcal{O}(X)} \mathrm{Vect}(X) = \Gamma(X, \mathrm{Sym}^n_{\mathcal{O}_X} T_X)$.

Recall that when X is smooth affine, we have a PBW filtration on $\mathcal{D}(X)$ by $\mathcal{O}(X)$-bimodules, and the PBW theorem asserts there are short exact sequences

$$0 \to \mathcal{D}_{n-1}(X) \to \mathcal{D}_n(X) \xrightarrow{\sigma_n} \mathrm{Sym}^n \mathrm{Vect}(X) \to 0.$$

Proposition 3.1. *Let X be smooth affine and irreducible, and let $f \in \mathcal{O}(X)$ be nonzero.*

1. *For all $u \in \mathcal{D}_n(X)$, there is a unique $u_f \in \mathcal{D}_n(X_f)$ such that the map $u_f : \mathcal{O}(X_f) \to \mathcal{O}(X_f)$ extends that of u.*
2. *$\sigma_n(u_f) = \sigma_n(u)|_{X_f}$.*

3. *The maps*

$$\mathcal{O}(X_f) \otimes_{\mathcal{O}(X)} \mathcal{D}_n(X) \to \mathcal{D}_n(X_f) \leftarrow \mathcal{D}_n(X) \otimes_{\mathcal{O}(X)} \mathcal{O}(X_f)$$

are isomorphisms.

Proof. We induct on n. For $n = 0$ the statement is clear check!. For $n = 1$ we use the fact that vector fields, i.e. derivations, can be extended to localizations via the quotient rule check in more detail!. For $n > 1$, we use linearity to reduce to the case where u is a product of vector fields. As in the $n = 1$ case, we extend u by extending each vector field. This gives existence, but we need uniqueness, since linear combinations are not unique. Suppose we have two extensions u_f, u'_f. Then $u_f - u'_f$ extends $0 \in \mathcal{D}_n(X)$. From the way we constructed extensions, we see that $\sigma_n(u_f) = \sigma_n(u)|_{X_f} = \sigma_n(u'_f)$. Then $u_f - u'_f \in \ker \sigma_n = \mathcal{D}_{n-1}(X_f)$, so by induction it is 0. I have a problem with this proof, which is when we use symbols. We don't know that any two extensions come from taking linear combinations... This shows (1) and (2). For (3), introduce the notation $\mathcal{O} = \mathcal{O}(X)$ and $\mathcal{O}_f = \mathcal{O}(X_f)$. The functor $\mathcal{O}_f \otimes_{\mathcal{O}} -$ is exact, so we have the following commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathcal{D}_{n-1}(X) \arrow[r] \arrow[d] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathcal{D}_n(X) \arrow[r] \arrow[d] & \mathcal{O}_f \otimes_{\mathcal{O}} \mathrm{Sym}^n \mathrm{Vect}(X) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{D}_{n-1}(X_f) \arrow[r] & \mathcal{D}_n(X_f) \arrow[r] & \mathrm{Sym}^n \mathrm{Vect}(X_f) \arrow[r] & 0
\end{tikzcd}$$

The left vertical arrow is an isomorphism by induction, and the right vertical arrow is an isomorphism by general algebraic geometry. Thus the middle vertical arrow is an isomorphism as desired.

As a consequence, for all smooth (irreducible) X, we have a quasicoherent sheaf of left/right $\mathcal{O}_X$ modules $\mathcal{D}_{X,n}$. Furthermore, we have a short exact sequence

$$0 \to \mathcal{D}_{X,n-1} \to \mathcal{D}_{X,n} \to \mathrm{Sym}^n T_X \to 0.$$

By induction on n, using the fact that $\mathcal{D}_{X,0} = \mathcal{O}_X$ and $\mathrm{Sym}^n T_X$ are coherent, we find that $\mathcal{D}_{X,n}$ is in fact coherent. We can then take $\mathcal{D}_X = \bigcup_n \mathcal{D}_{X,n}$, a quasicoherent sheaf of associative algebras. PBW implies $\mathrm{gr}\mathcal{D}_X \cong \mathrm{Sym} T_X$.

3.2 Twisted differential operators (TDO)

Let $\mathcal{L}$ be a line bundle on X. Let $\mathcal{D}_{X,\mathcal{L}} = \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{D}_X \otimes_{\mathcal{O}_X} \mathcal{L}^{-1}$. Here $\mathcal{L}^{-1}$ is the dual bundle to $\mathcal{L}$, sometimes denoted $\mathcal{L}^\vee$ or $\mathcal{L}^*$.

Claim 3.1. *$\mathcal{D}_{X,L}$ has the structure of a sheaf of associative algebras.*

Proof. Let u_1, u_2 be local sections of $\mathcal{D}_X$. Let α_1, α_2 be local sections of $\mathcal{L}$. Let β_1, β_2 be local sections of $\mathcal{L}^{-1}$. We define $(\alpha_1 u_1 \beta_1)(\alpha_2 u_2 \beta_2)$ to be $\alpha_1(u_1 m_{\langle \beta_1, \alpha_2 \rangle} u_2)\beta_2$.

There is also a natural action of $\mathcal{D}_{X,\mathcal{L}}$ on $\mathcal{L}$, given on local sections by $(\alpha_1 u\beta)\alpha_2 = u(\langle\beta,\alpha_2\rangle)\alpha_1$.

Replacing $\mathcal{D}_X$ by $\mathcal{D}_{X,n}$ in the definition of $\mathcal{D}_{X,\mathcal{L}}$ gives a sheaf we call $\mathcal{D}_{X,\mathcal{L},n}$, giving a PBW filtration on $\mathcal{D}_{X,\mathcal{L}}$. We have

$$\mathrm{gr}\mathcal{D}_{X,\mathcal{L}} = \mathcal{L} \otimes_{\mathcal{O}_X} \mathrm{gr}\mathcal{D}_X \otimes_{\mathcal{O}_X} \mathcal{L}^{-1} = \mathrm{gr}\mathcal{D}_X,$$

where we can cancel $\mathcal{L}$ and $\mathcal{L}^{-1}$ since $\mathrm{gr}\mathcal{D}_X$ is an ordinary sheaf of $\mathcal{O}_X$ modules, i.e. there is no distinction between left and right.

Definition 3.1. A $\mathcal{D}_X$**-module** is a quasicoherent sheaf $\mathcal{M}$ such that for all open $U \subset X$, we have an action

$$\Gamma(U,\mathcal{D}_X) \times \Gamma(U,\mathcal{M}) \to \Gamma(U,\mathcal{M}).$$

Recall $\mathcal{D}_X = \mathcal{U}(\mathcal{O}_X + T_X)/(f \otimes u = fu \forall f \in \mathcal{O}_X, u \in \mathcal{D}_{X,1}; 1_{\mathcal{U}} = 1_{\mathcal{O}_X})$. Thus, a $\mathcal{D}_X$-module is the same as a quasicoherent sheaf $\mathcal{M}$ with a "flat connection", i.e. maps ∇_ξ for $\xi \in T_X$ acting on local sections of $\mathcal{M}$ such that $\nabla_\xi(fm) = f\nabla_\xi m + \xi(f)m$ and $\nabla_{[\xi,\eta]} = [\nabla_\xi, \nabla_\eta]$.

For a $\mathcal{D}_X$-module $\mathcal{M}$, $\Gamma(X,\mathcal{M})$ is a module over $\Gamma(X,\mathcal{D}_X) = \mathcal{D}(X)$. In particular, $\Gamma(X,-)$ gives a functor from $\mathcal{D}_X$-modules to $\mathcal{D}(X)$-modules. It has a left adjoint called the localization functor Loc, given by $\mathrm{Loc}(M) = \mathcal{D}_X \otimes_{\mathcal{D}(X)} M$. In particular, for an affine open $U \subset X$, we have $\Gamma(U,\mathrm{Loc}(M)) = \mathcal{D}(U) \otimes_{\mathcal{D}(X)} M$.

If X is affine, $\Gamma(X,-)$ is an equivalence $\mathrm{QCoh}(X) \to \mathcal{O}(X)$-mod with inverse $M \mapsto \mathcal{O}_X \otimes_{\mathcal{O}(X)} M$. Using the map 1, one can show that this restricted map from $\mathcal{D}_X$-modules to $\mathcal{D}(X)$-modules is also an equivalence (when X is affine).

3.3 Beilinson-Bernstein localization theorem, the statement

Let G be a connected semisimple algebraic group with fixed Borel subgroup B. Let $\mathcal{B} = G/B$ be the flag variety.

Theorem 3.1 (Beilinson-Bernstein Localization Theorem)**.** $\Gamma(\mathcal{B},-) : \mathcal{D}_{\mathcal{B}}$-mod $\to$ $\mathcal{D}(\mathcal{B})$-mod *is an equivalence with inverse* Loc.

Remark. A similar statement holds for all partial flag varieties G/P with P parabolic. In particular, it holds for projective spaces. Historically, this statement was first shown for projective spaces in a much simpler way.

4 Apr 2

4.1 Beilinson-Bernstein localization theorem, the proof (part 1)

We start the proof of Theorem 3.1.

4.1.1 Setup

Let G be a connected semisimple algebraic group with fixed Borel subgroup B_0. Let $\mathcal{B} = G/B_0$ be the flag variety. Let D denote the category of $\mathcal{D}_{\mathcal{B}}$-modules. We will often abbreviate $\Gamma(\mathcal{B}, -)$ to just $\Gamma(-)$.

4.1.2 Today's goal

There are two key statements:

1. If $0 \neq \mathcal{M} \in \mathsf{D}$, then $\Gamma(\mathcal{M}) \neq 0$.
2. For all $\mathcal{M} \in \mathsf{D}$ and $i > 0$, we have $H^i(\mathcal{M}) = 0$.

These statements imply that Γ is exact on D. We will then use this to show that any $\mathcal{M} \in \mathsf{D}$ is generated by global sections.

4.1.3 Step 1

Fix a maximal torus $T \subset B_0$. We use $\lambda \in X^*(T) = \mathrm{Hom}(T, \mathbb{C}^\times)$ to denote a character of T.

Fix V, a finite dimensional irreducible representation of G. There is a unique 1-dimensional subspace $\ell(B_0) \subset V$ called the highest weight space such that B_0 acts on $\ell(B_0)$ via a dominant character λ.

Note that G/B_0 can be identified with the set of Borels in G, via $gB_0 \mapsto B = gB_0g^{-1}$. Set $\ell(B) = g\ell(B_0)$. The collection of lines $\ell(B)$ give a line bundle on $\mathcal{B}$. We let $\mathcal{O}(\lambda)$ be the corresponding coherent sheaf of local sections. In particular, there is a map $\iota : \mathcal{B} \to \mathbb{P}(V)$ given by $B \mapsto \ell(B)$, and $\mathcal{O}(\lambda) = \iota^*\mathcal{O}(-1)$. We say λ is **regular** if ι is an embedding; in this case, $\mathcal{O}(-\lambda) = \iota^*\mathcal{O}(1)$ is very ample.

4.1.4 Step 2

By Lie's theorem, we may choose a B_0-stable flag in V starting with $\ell(B_0)$, i.e. subspaces $\ell(B_0) = F_1 \subset F_2 \subset \cdots \subset F_n = V$ where each F_i is B_0-stable and $\dim F_i = i$. Then B_0 acts on F_i/F_{i-1} by a character $\lambda_i \in X^*(T)$. The characters λ_i are the weights of V. We let λ^* denote λ_n; it is the lowest weight of V.

For $B = gB_0g^{-1}$ another Borel, we define $F_i(B) = gF_i$. Then $F_\bullet(B)$ is a B-stable flag in V. For a fixed i, the i-dimensional spaces $F_i(B)$ for $B \in \mathcal{B}$ give a rank i vector bundle on $\mathcal{B}$, which we denote by $F_i(\mathcal{B})$. We obtain a flag of subvector bundles of $F_n(\mathcal{B}) = V \times \mathcal{B}$. We let $\mathcal{F}_i(\mathcal{B})$ denote the corresponding locally free sheaf to each $F_i(\mathcal{B})$. In particular, $\mathcal{F}_1(\mathcal{B}) = \mathcal{O}(\lambda)$ and $\mathcal{F}_n(\mathcal{B}) = V \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{B}}$. We note that $F_i(\mathcal{B})/F_{i-1}(\mathcal{B})$ is a line bundle whose corresponding sheaf is $\mathcal{O}(\lambda_i)$. Note that we did not define $\mathcal{O}(\nu)$ for an arbitrary character ν, but we will do this next time.

4.1.5 Step 3

Let $\mathcal{M} \in \mathsf{D}$. For any $\nu \in X^*(T)$, let $\mathcal{M}(\nu) = \mathcal{O}(\nu) \otimes_{\mathcal{O}_{\mathcal{B}}} \mathcal{M}$. This is a quasicoherent $\mathcal{O}_{\mathcal{B}}$-module. We also define $\mathcal{F}_i(\mu) = \mathcal{F}_i(\mathcal{B}) \otimes_{\mathcal{O}_{\mathcal{B}}} \mathcal{M}$. We get a flag $\mathcal{M}(\lambda) = \mathcal{F}_1(\mathcal{M}) \subset \cdots \subset \mathcal{F}_n(\mathcal{M}) = V \otimes_{\mathbb{C}} \mathcal{M}$. Since the $\mathcal{F}_i(\mathcal{B})$ are locally free, tensoring is exact, so $\mathcal{F}_i(\mathcal{M})/\mathcal{F}_{i-1}(\mathcal{M}) = \mathcal{M}(\lambda_i)$.

We now have two important short exact sequences. The first is

$$0 \to \mathcal{F}_{n-1}(\mathcal{M}) \to V \otimes_{\mathbb{C}} \mathcal{M} \to \mathcal{M}(\lambda^*) \to 0. \tag{2}$$

For the second, we first note that $V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)$ has a filtration by $\mathcal{F}_i(\mathcal{M})(-\lambda)$, and $\mathcal{F}_1(\mathcal{M})(-\lambda) = \mathcal{M}$. The second short exact sequence is

$$0 \to \mathcal{M} \to V \otimes_{\mathbb{C}} \mu(-\lambda) \to \mathcal{F}_n(\mathcal{M})(-\lambda)/\mathcal{M} \to 0. \tag{3}$$

4.1.6 Key lemma

Note that, so far, we have not used that $\mathcal{M}$ is a D-module. The following lemma is where this is used.

Lemma 4.1 (Key Lemma)**.** *If $\mathcal{M} \in \mathsf{D}$, then the short exact sequences 2 and 3 split as sheaves of $\mathbb{C}$-vector spaces on $\mathcal{B}$.*

We will prove the lemma next time.

4.1.7 Step 4

We now show that higher cohomology of $\mathcal{M} \in \mathsf{D}$ vanishes.

Fix $i > 0$. Let $b : \mathcal{N} \hookrightarrow \mathcal{M}$ be an embedding of a coherent subsheaf. All the filtration business from before can be done with $\mathcal{N}$, since we never used the D-module structure. We have a commuting square:

$$\begin{tikzcd}
\mathcal{N} \arrow[r, "a"] \arrow[d, hook, "b"'] & V \otimes_{\mathbb{C}} \mathcal{N}(-\lambda) \arrow[d, "\widetilde{b}"] \\
\mathcal{M} \arrow[r, "\widetilde{a}"'] & V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)
\end{tikzcd}$$

This induces a commuting square on cohomology:

$$\begin{tikzcd}
H^i(\mathcal{N}) \arrow[r, "H^i(a)"] \arrow[d, hook, "H^i(b)"'] & H^i(V \otimes_{\mathbb{C}} \mathcal{N}(-\lambda)) \arrow[r, equal] \arrow[d, "H^i(\widetilde{b})"] & V \otimes_{\mathbb{C}} H^i(N(-\lambda)) \\
H^i(\mathcal{M}) \arrow[r, "H^i(\widetilde{a})"'] & H^i(V \otimes_{\mathbb{C}} \mathcal{M}(-\lambda)) \arrow[r, equal] & V \otimes_{\mathbb{C}} H^i(M(-\lambda))
\end{tikzcd}$$

Now we choose λ large enough so that $\mathcal{O}(-\lambda)$ is very ample and $H^i(\mathcal{N}(-\lambda)) = 0$. Thus $H^i(\widetilde{a}) \circ H^i(b) = 0$. By Lemma 4.1, $\widetilde{a}$ is an embedding of a direct summand, so that $H^i(\widetilde{a})$ is injective. Thus $H^i(b) = 0$.

We have shown that for any coherent subsheaf $\mathcal{N} \subset \mathcal{M}$, the induced map $H^i(\mathcal{N}) \to H^i(\mathcal{M})$ is 0. But $\mathcal{M}$ is a direct limit of its coherent subsheaves, and H^i commutes with direct limits, so we find $H^i(\mathcal{M}) = 0$.

4.1.8 Step 5

We now show that the global sections of a nonzero $\mathcal{M} \in \mathsf{D}$ is nonzero.

There is a nonzero coherent subsheaf $\mathcal{N} \subset \mathcal{M}$. Choose λ such that $\mathcal{O}(\lambda^*)$ is very ample and $\Gamma(\mathcal{N}(\lambda^*)) \neq 0$. Then the injection $\Gamma(\mathcal{N}(\lambda^*)) \hookrightarrow \Gamma(\mathcal{M}(\lambda^*))$ shows that $\Gamma(\mathcal{M}(\lambda^*))$. By Lemma 4.1, we also have an injection $\Gamma(\mathcal{M}(\lambda^*)) \hookrightarrow \Gamma(V \otimes_{\mathbb{C}} \mathcal{M}) = V \otimes_{\mathbb{C}} \Gamma(\mathcal{M})$. Thus $\Gamma(\mathcal{M}) \neq 0$ as desired.

4.1.9 Step 6

We now show that any $\mathcal{M} \in \mathsf{D}$ is generated by global sections.

Let $M = \Gamma(\mathcal{M})$; it is a $\mathcal{D}(\mathcal{B})$-module. Recall that Γ has a left adjoint $\mathrm{Loc} : N \mapsto \mathcal{D}_{\mathcal{B}} \otimes_{\mathcal{D}(\mathcal{B})} N$. The adjunction gives a map $\alpha : \mathrm{Loc}(M) \to \mathcal{M}$. By definition, $\mathcal{M}$ is generated by global sections means that α is surjective. Let $\mathcal{M}'$ be the image of α. The identity $M \to M$ factors as

$$M \xrightarrow{m \mapsto 1 \otimes m} \Gamma(\mathrm{Loc}(M)) \xrightarrow{\Gamma(\alpha)} \Gamma(\mathcal{M}') \hookrightarrow \Gamma(\mathcal{M}) = M.$$

Thus implies that $\Gamma(\mathcal{M}') \xrightarrow{\sim} \Gamma(\mathcal{M})$. Now let $\mathcal{M}''$ be the cokernel of α, so we have a short exact sequence

$$0 \to \mathcal{M}' \to \mathcal{M} \to \mathcal{M}'' \to 0.$$

From steps 4 and 5, we know $\Gamma(-)$ is exact, so we get an exact sequence

$$0 \to \Gamma(\mathcal{M}') \to \Gamma(\mathcal{M}) \to \Gamma(\mathcal{M}'') \to 0.$$

But the map $\Gamma(\mathcal{M}') \to \Gamma(\mathcal{M})$ is an isomorphism, so $\Gamma(\mathcal{M}'') = 0$. By step 5, this means that $\mathcal{M}'' = 0$, as desired.

4.2 Some setup for next time

Let an algebraic group G act on a variety X. Let $E \xrightarrow{p} X$ be a vector bundle. We say the bundle is G-equivariant if

1. G acts on E so that p is an equivariant map.

2. The induced maps $E_x \to E_{gx}$ on fibers are linear.

Let $\mathcal{E}$ be the sheaf of sections of E, and let $\mathfrak{g} = \mathrm{Lie}G$. There is a Lie derivative map which associates to $\xi \in \mathfrak{g}$ a map $\nabla_\xi : \mathcal{E} \to \mathcal{E}$. In the analytic setting, $\nabla_\xi(s) = \frac{d}{dt}(e^{-t\xi}(s))|_{t=0}$. There is an algebraic way to make sense of this as well.